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Equivalent Circuit (EC) FDTD Method for Dispersive Materials: Derivation, Stability Criteria and Application Examples

A. Rennings, A. Lauer, C. Caloz and I. Wolff

1 Introduction

Although many computational methods are currently available to analyze electromagnetic problems of high complexity, further progress is necessary to unify the different approaches and to improve the robustness of some algorithms. This chapter aims at reaching these two goals. It presents an equivalent circuit (EC) finite difference time domain (FDTD) method which both represents a complete circuitual formulation of conventional FDTD [1] based on the electric and magnetic fields Yee scheme and provides guaranteed stability criteria for dispersive media.

A circuitual formulation of FDTD offers several benefits. Since its update values are voltages and currents (as opposed to electric and magnetic fields), it facilitates the integration of electronic components, such as for instance diodes and transistors, into the simulation. In addition, it provides increased computational speed thanks to a reduced number of multiplications in the update equations [2, 3]. Finally, it may be easily generalized to arbitrary dispersive media via simple manipulations of the mesh immittances.

While no convenient explicit guaranteed stability criterion for dispersive media is available in the framework of the conventional FDTD technique, the EC FDTD provides such a criterion. This is a distinct advantage for both emerging metamaterials [4], which are inherently dispersive, and for various natural materials [5], such as metals at the optical frequencies, ferrimagnetic and ferromagnetic materials and biological tissues, which exhibit various types of more or less complex dispersion responses.

A. Rennings

IMST GmbH, D-47475 Kamp-Lintfort, Germany, e-mail: A.Rennings@ieee.org

A. Lauer

IMST GmbH, D-47475 Kamp-Lintfort, Germany

C. Caloz

École Polytechnique, Montréal, H3T 1J4, Québec, Canada, e-mail: christophe.caloz@polymtl.ca

I. Wolff

IMST GmbH, D-47475 Kamp-Lintfort, Germany

The chapter is organized as follows. Section 2 introduces a compact and convenient notation for the method, derives the EC FDTD method first for the simplest case of non-dispersive materials and then extends it for the case of dispersive materials, including the cases of Drude, Lorentz and Debye media. Section 3 develops within the framework of the EC FDTD a discretized and circuital version of the Poynting theorem both for the time-continuous and time-discrete cases. Upon this foundation and following a Lyapunov energy approach, Sect. 4 establishes a guaranteed stability criterium, which may be particularized to arbitrary types of dispersion responses. Application examples of metamaterial and natural material dispersive structures and devices are next presented in Sect. 5. Finally, Sect. 6 outlines the main results.

2 Derivation of the Method

2.1 Notation of the Discrete Calculus

Here a compact and convenient notation for the EC FDTD method is introduced. Additionally to conventional mathematical notations here all operators including matrices are denoted by upright characters (like $S[\cdot]$), and vectorial/tensorial objects are denoted by capital bold characters (vector field $\mathbf{E}$ or matrix $\mathbf{A}$).

2.1.1 Discrete Fields

Each of the six continuous scalar fields of the form $F_{\text{conti}}(\mathbf{x}, t)$, $\mathbf{x} \in \mathbb{R}^3$, $t \in \mathbb{R}$ of the electromagnetic field are represented by six *discrete fields* of the form $F_{\text{disc}}^n(\mathbf{X})$, $\mathbf{X} \in \mathbb{Z}^3$, $n \in \mathbb{Z}$, where $\mathbf{X}$ is the three-dimensional spatial index vector and n is the time index. The index vector $\mathbf{X}$ is usually not given [F_{disc}^n and not $F_{\text{disc}}^n(\mathbf{X})$], since it is assumed that all discrete fields depend on $\mathbf{X}$. Whereas the time index n is always given for the time discrete case (conventional technique with upper index).

2.1.2 Modulo-Three Notation and Variable Indices

For the sake of simpler and compacter notation, component and spatial direction indices are numbered in a modulo-three sense, e.g. $F_0 \equiv F_x$, $F_1 \equiv F_y$, $F_2 \equiv F_z$ and $F_3 \equiv F_0$, $F_4 \equiv F_1$, i.e., the effective index is always element of the set $\{0, 1, 2\}$.

In 3D figures the direction indices $d \in \{0, 1, 2\}$ are explicitly given (as absolute numbers), whereas in 2D figures (projections of a 3D arrangement along the Cartesian axes) variable indices ($d, d+1, d+2$) are used. By inserting $\{0, 1, 2\}$ for d one gets the projection onto a desired plane. The variable indices are also utilized to specify vectorial equations in a scalar manner. Again, by inserting $\{0, 1, 2\}$ for d the

specific component is obtained. The fact that d is element of the set $\{0, 1, 2\}$ is not repeated in each equation with variable indices, it is part of the general assumptions outlined in this section.

2.1.3 Discrete Spatial-Shift and Index-Shift Operators

In finite difference schemes the application of spatial operators onto discrete fields is usually specified by using index changes together with appropriate weighting factors. Quite often the resulting equations are cumbersome. In order to circumvent this, we introduce shift operators S_d and its inverse S_d^{-1} . Together with the null $0[\cdot]$ and the identity element $I[\cdot]$ the family of shift operators generate an operator space $\mathbb{O}$, which in algebraic terms is called a commutative ring. The shift operators are defined via their action onto the discrete fields. It is

$$S_d F(\mathbf{X}) = F(\mathbf{X} + \mathbf{E}_d), \quad \{S_d^{-1} + S_{d+1}\} F(\mathbf{X}) = F(\mathbf{X} - \mathbf{E}_d) + F(\mathbf{X} + \mathbf{E}_{d+1}), \quad (1)$$

with the index unit vectors $\mathbf{E}_0 = (1, 0, 0)^T$, $\mathbf{E}_1 = (0, 1, 0)^T$ and $\mathbf{E}_2 = (0, 0, 1)^T$. In Eq. (1) the summation in $\mathbb{O}$ is also shown. If a summation of several operators is applied to a field, braces $\{\dots\}$ are used around the operator sum as in Eq. (1). The application of S_d to a field that does not depend on X_d yields the unchanged field (identity operation). This rule is important for the application of shift-operators to fields that depend only on one component of the index vector, like the dimensions of the mesh cells. For instance Δ_d is only a function of X_d . Thus expressions like $S_d \Delta_d$ can be simplified to $S \Delta_d$.

The following example should clarify the multiplication in $\mathbb{O}$:

$$\alpha S_d S_{d+2}^{-1} F(\mathbf{X}) = S_{d+2}^{-1} S_d \alpha F(\mathbf{X}) = \alpha F(\mathbf{X} + \mathbf{E}_d - \mathbf{E}_{d+2}) \quad (2)$$

Due to inhomogeneous material distributions the simple multiplication with a constant factor in $\mathbb{O}$ needs to be extended to a multiplicative operator $M(\mathbf{X})$ that depends on $\mathbf{X}$. In this case the commutativity of the multiplication in $\mathbb{O}$ is lost, since e.g.

$$S_d [M(\mathbf{X}) F(\mathbf{X})] = M(\mathbf{X} + \mathbf{E}_d) F(\mathbf{X} + \mathbf{E}_d) \neq M(\mathbf{X}) S_d F(\mathbf{X}). \quad (3)$$

If an operator is applied to more than the field directly behind it, brackets $[\dots]$ are used, for instance $S_d [M(\mathbf{X}) F_d(\mathbf{X}) + F_{d+2}(\mathbf{X})]$.

The index-shift operator R and its inverse R^{-1} are defined by

$$R F_d = F_{d+1}, \quad R^{-1} F_d = F_{d-1}. \quad (4)$$

This operator is used in Sect. 3 where a discrete Poynting theorem will be derived. R is used for re-formulations and simplification of terms that are summed up over all three directions $d \in \{0, 1, 2\}$. The overall sum does not change if these operators are applied due to the modulo-3 sense of the indices.

2.2 Basic EC FDTD for Non-Dispersive Materials

The basic EC FDTD scheme is derived from an approximation of Maxwell's equations in integral form, including electric loss σ_e , magnetic loss σ_m , and a current density source $\mathbf{J}_{\text{src}}$ (soft excitation),

$$\oint_{\partial A} \mathbf{H} \cdot d\boldsymbol{\ell} = + \frac{d}{dt} \iint_A \varepsilon \mathbf{E} \cdot d\mathbf{A} + \iint_A \sigma_e \mathbf{E} \cdot d\mathbf{A} - \iint_A \mathbf{J}_{\text{src}} \cdot d\mathbf{A}, \quad (5)$$

$$\oint_{\partial A} \mathbf{E} \cdot d\boldsymbol{\ell} = - \frac{d}{dt} \iint_A \mu \mathbf{H} \cdot d\mathbf{A} - \iint_A \sigma_m \mathbf{H} \cdot d\mathbf{A}, \quad (6)$$

where all material constitutive parameters ($\varepsilon, \mu, \sigma_e, \sigma_m$) are considered *non-dispersive*, i.e. independent of frequency (but possibly inhomogeneous).

These space-time continuous equations are *spatially discretized* using the well-known *Yee cell* [1], which is depicted in Fig. 1. Within the Yee cell, the electric field is sampled at the edges of the main mesh subcell while the magnetic field is sampled at the edges of the dual mesh subcell. The field samples within the Yee cell are located by the index vector $\mathbf{X} = (X_0, X_1, X_2)^T$, according to the notations of Sect. 2.1.

The Yee discretization details of the proposed EC FDTD formulation are shown in Fig. 2. The magnetic field samples are located normally at the *center* of the faces of the main mesh subcells and are *symmetrically surrounded* by four adjacent tangential electric field samples. In contrast, in the general case of a non-uniform mesh, the electric field samples are *not* located symmetrically with respect to the surrounding magnetic field samples.

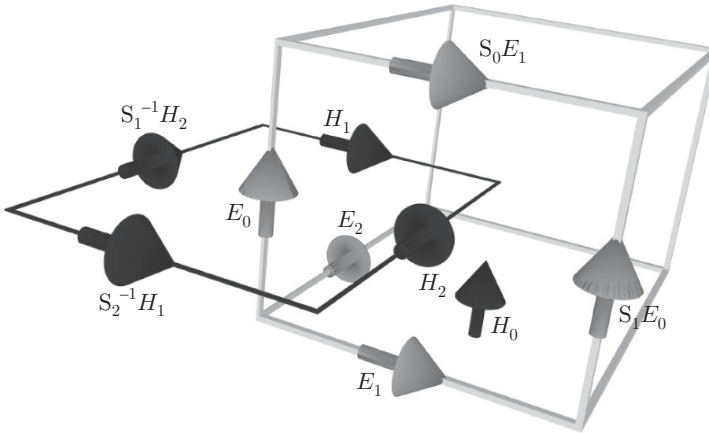


Fig. 1 Yee cell, which consists of a *main mesh subcell* (*parallelepipedic structure* in the figure) and a *dual mesh subcell* (*staggered parallelepipedic structure*, of which only one face is shown in the figure). The Yee cell includes six unshifted field samples. Additionally four positively or negatively shifted field samples are shown, which are required for the approximation of Eqs. (5) and (6)

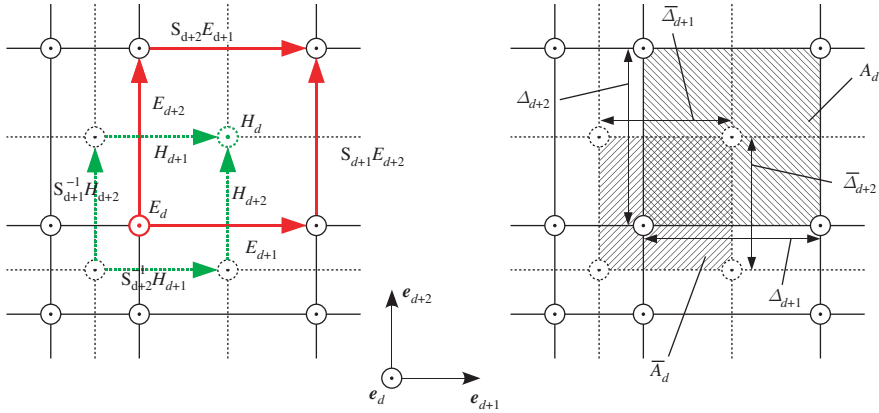


Fig. 2 Details of the Yee cell (Fig. 1) discretization. Used components and corresponding subcell dimensions for the spatial discretization of Eqs. (5) and (6) leading to Eqs. (8) and (9)

Here in this scheme the electric *and also* the magnetic material parameters are approximated by *main mesh “blocks” with piecewise constant constitutive parameters*¹, which allows a natural meshing of inhomogeneous materials, since a change in the material parameters is mostly due to an object boundary. In the case of a non-uniform mesh, the dimensions of the mesh cells vary across space, and the edge length Δ_d is only a function of the component X_d of the index vector $\mathbf{X}$. According to the notations of Sect. 2.1, the dimensions of the dual mesh subcell, $\bar{\Delta}_d$, are related to the dimensions of the main mesh subcell, Δ_d , by

$$\bar{\Delta}_d = \frac{1}{2} (S_d^{-1} \Delta_d + \Delta_d) = \frac{1}{2} \{S_d^{-1} + 1\} \Delta_d, \quad \forall \quad d \in \{0, 1, 2\}. \quad (7)$$

Evaluating Maxwell's Eqs. (5) and (6) on all faces of the main and dual mesh subcells, as represented in Fig. 2, yields the (still time-continuous) spatially discretized generic equations

$$\oint_{\partial \bar{A}_d} \mathbf{H} \cdot d\mathbf{l} \approx \{1 - S_{d+1}^{-1}\} H_{d+2} \bar{\Delta}_{d+2} - \{1 - S_{d+2}^{-1}\} H_{d+1} \bar{\Delta}_{d+1} = \quad (8)$$

$$+ \left(\epsilon_{d,\text{eff}} \frac{d}{dt} E_d + \sigma_{d,\text{eff}}^e E_d - J_d^{\text{src}} \right) \bar{\Delta}_{d+1} \bar{\Delta}_{d+2} \approx \iint_{\bar{A}_d} \dots d\mathbf{A},$$

$$\oint_{\partial A_d} \mathbf{E} \cdot d\mathbf{l} \approx \{S_{d+1} - 1\} E_{d+2} \Delta_{d+2} - \{S_{d+2} - 1\} E_{d+1} \Delta_{d+1} = \quad (9)$$

$$- \left(\mu_{d,\text{eff}} \frac{d}{dt} H_d + \sigma_{d,\text{eff}}^m H_d \right) \Delta_{d+1} \Delta_{d+2} \approx \iint_{A_d} \dots d\mathbf{A},$$

¹ Usually the magnetic material parameters are approximated by *dual* mesh blocks with piecewise constant constitutive parameters.

where the four effective constitutive parameters $\epsilon_{d,\text{eff}}$, $\mu_{d,\text{eff}}$, $\sigma_{d,\text{eff}}^e$, $\sigma_{d,\text{eff}}^m$ will be determined now.

The effective *electric* parameters are approximated by averaging the fluxes $\iint_{\bar{A}_d} \{\epsilon, \sigma_e\} \mathbf{E} \cdot d\mathbf{A}$ on each face $\bar{A}_d$ (Fig. 2) of the *dual* mesh subcells, leading to

$$\epsilon_{d,\text{eff}} \doteq \frac{\frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\epsilon \Delta_{d+1} \Delta_{d+2}]}{\frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\Delta_{d+1} \Delta_{d+2}]}, \quad (10)$$

for $\epsilon_{d,\text{eff}}$, as illustrated in Fig. 3(a), and $\sigma_{d,\text{eff}}^e$ is evaluated in the same manner.

The magnetic parameters are constant in each A_d -transversal direction (where A_d is defined by the directions $d+1$ and $d+2$, Fig. 2, right) of the *main* mesh subcells, and therefore no useful averaging quantity may be defined from the fluxes $\iint_{A_d} \{\mu, \sigma_m\} \mathbf{H} \cdot d\mathbf{A}$. In contrast, two different magnetic materials may exist in the longitudinal direction (d), and therefore the effective *magnetic* parameters may be approximated by averaging the magneto-motive force² $H_d \bar{\Delta}_d$ as illustrated in Fig. 3(b)

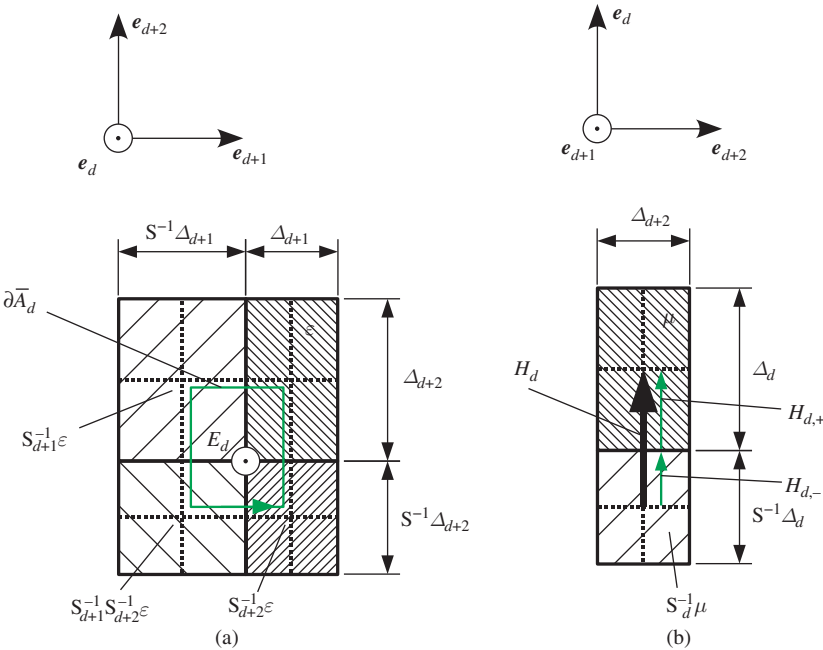


Fig. 3 Averaging process for the approximation of the effective constitutive parameters. (a) Local virtual capacitance filled with 4 different materials to define $\epsilon_{d,\text{eff}}$. (b) Local virtual inductance filled with 2 different materials to define $\mu_{d,\text{eff}}$

² The magnetic flux density B_d is continuous here.

$$H_d \bar{\Delta}_d = H_{d,+} \frac{1}{2} \Delta_d + H_{d,-} \frac{1}{2} S_d^{-1} \Delta_d = B_d \frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\mu} \right] \doteq B_d \frac{\bar{\Delta}_d}{\mu_{d,\text{eff}}}, \quad (11)$$

from which

$$\mu_{d,\text{eff}} \doteq \frac{\bar{\Delta}_d}{\frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\mu} \right]} = \frac{\frac{1}{2} \{1 + S_d^{-1}\} \Delta_d}{\frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\mu} \right]}, \quad (12)$$

for $\mu_{d,\text{eff}}$, and $\sigma_{d,\text{eff}}^m$ is evaluated in the same manner.

Inserting the so-defined effective constitutive parameters into Eqs. (8) and (9) yields the complete space-discretized Maxwell's equations

$$\begin{aligned} \{1 - S_{d+1}^{-1}\} H_{d+2} \bar{\Delta}_{d+2} - \{1 - S_{d+2}^{-1}\} H_{d+1} \bar{\Delta}_{d+1} &= -\bar{\Delta}_{d+1} \bar{\Delta}_{d+2} J_d^{\text{src}} \\ &+ \frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\epsilon \Delta_{d+1} \Delta_{d+2}] \frac{1}{\Delta_d} \frac{d}{dt} E_d \Delta_d \\ &+ \frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\sigma_e \Delta_{d+1} \Delta_{d+2}] \frac{1}{\Delta_d} E_d \Delta_d, \end{aligned} \quad (13)$$

$$\begin{aligned} \{S_{d+1} - 1\} E_{d+2} \Delta_{d+2} - \{S_{d+2} - 1\} E_{d+1} \Delta_{d+1} &= \\ - \frac{\Delta_{d+1} \Delta_{d+2}}{\frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\mu} \right]} \frac{d}{dt} H_d \bar{\Delta}_d + \frac{\Delta_{d+1} \Delta_{d+2}}{\frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\sigma_m} \right]} H_d \bar{\Delta}_d, \end{aligned} \quad (14)$$

where the terms $E_d \Delta_d \doteq v_d$ may be interpreted as the electric voltages along the edges of the main mesh subcells and the terms $H_d \bar{\Delta}_d \doteq i_d$ may be interpreted as the “magnetic voltages” along the edges of the dual mesh subcells.³

This circuit analogy and inspection of Eqs. (13) and (14) naturally lead to the definition of the equivalent *lumped elements*

$$C_d = \frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\epsilon \Delta_{d+1} \Delta_{d+2}] \frac{1}{\Delta_d}, \quad (15)$$

$$G_d = \frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\sigma_e \Delta_{d+1} \Delta_{d+2}] \frac{1}{\Delta_d}, \quad (16)$$

which are interpreted as the local capacitance and conductance, and

$$L_d = \frac{\Delta_{d+1} \Delta_{d+2}}{\frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\mu} \right]}, \quad R_d = \frac{\Delta_{d+1} \Delta_{d+2}}{\frac{1}{2} \{1 + S_d^{-1}\} \left[\frac{\Delta_d}{\sigma_m} \right]}, \quad (17)$$

which are interpreted as the local inductance and resistance. To complete the circuit analogy, a lumped current source i_d^{src} is defined

³ Alternatively the term $H_d \bar{\Delta}_d \doteq i_d$ can be interpreted as corresponding loop current in the face A_d of the main mesh cell (Fig. 4), which is related to the orthogonal magnetic field H_d component by the right-hand rule.

$$i_d^{\text{src}} = \frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\Delta_{d+1} \Delta_{d+2}] J_d^{\text{src}}. \quad (18)$$

With these local lumped elements Ampere's (5) and Faraday's (6) laws take the very compact form

$$\{1 - S_{d+1}^{-1}\} i_{d+2} - \{1 - S_{d+2}^{-1}\} i_{d+1} + i_d^{\text{src}} = +C_d \frac{d}{dt} v_d + G_d v_d, \quad (19)$$

$$\{S_{d+1} - 1\} v_{d+2} - \{S_{d+2} - 1\} v_{d+1} = -L_d \frac{d}{dt} i_d - R_d i_d. \quad (20)$$

For each $d \in \{0, 1, 2\}$ Eq. (19) represents the Kirchhoff's current law for a current node and Eq. (20) represents the Kirchhoff's voltage law for a voltage loop. This "circuitization" of the Yee FDTD discretization of space may be represented by the equivalent circuit cell shown in Fig. 4.

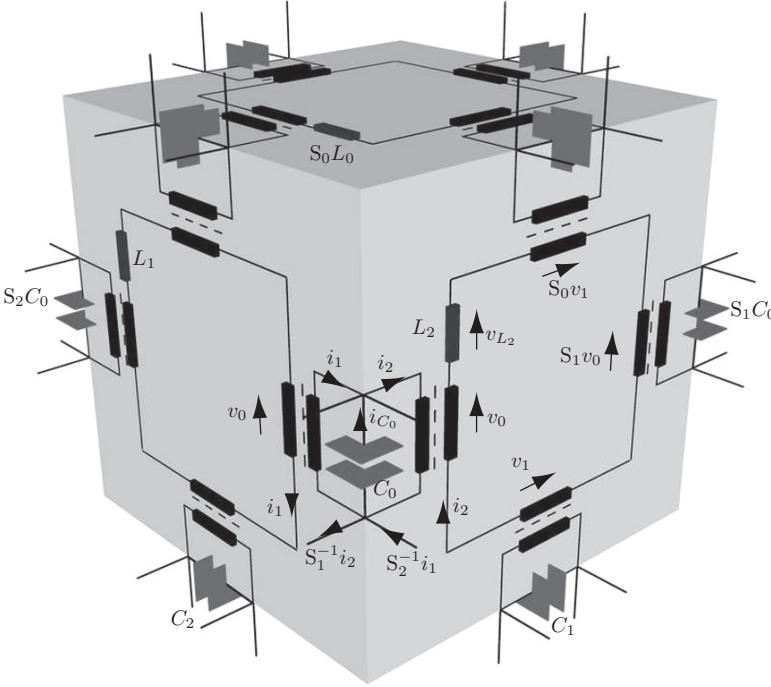


Fig. 4 3D equivalent circuit cell for a non-dispersive material with (*shunt*) capacitances located at the edges of the main mesh cells (*e-nodes*) and (*series*) inductances (represented by the *rectangle in the loop*) located at the faces of the main mesh cells. The 1:1-transformers are necessary to ensure a well defined loop current with a *one-to-one* correspondence to the orthogonal magnetic field component. The direction of the voltages v_d is positive in the parallel Cartesian direction e_d and the one for the loop currents i_d is positive with respect to the corresponding magnetic field component (*right-hand rule*). The values $i_{C_0} = C_0 \frac{d}{dt} v_0$ and $v_{L_2} = L_2 \frac{d}{dt} i_2$ are used as an abbreviation within the figure

The final step in the derivation of the EC FDTD scheme consists in *time-discretize* by finite difference the previous equations in order to set up the update equations. Defining a time step Δt and using the traditional leap-frog scheme, the voltage values are sampled at time multiples of Δt while the current values are sampled between these instances, i.e. at odd multiples of $\Delta t/2$. This leads to

$$\{1 - S_{d+1}^{-1}\} i_{d+2}^{n-\frac{1}{2}} - \{1 - S_{d+2}^{-1}\} i_{d+1}^{n-\frac{1}{2}} = +C_d \frac{v_d^n - v_d^{n-1}}{\Delta t} + G_d \frac{v_d^n + v_d^{n-1}}{2} - i_{\text{src},d}^{n-\frac{1}{2}}, \quad (21)$$

$$\{S_{d+1} - 1\} v_{d+2}^n - \{S_{d+2} - 1\} v_{d+1}^n = -L_d \frac{i_d^{n+\frac{1}{2}} - i_d^{n-\frac{1}{2}}}{\Delta t} - R_d \frac{i_d^{n+\frac{1}{2}} + i_d^{n-\frac{1}{2}}}{2}, \quad (22)$$

and solving these equations for the latest voltages v_d^n and currents $i_d^{n+\frac{1}{2}}$ yield the final EC FDTD update equations for a non-dispersive material region of the computational domain

$$v_d^n = \frac{1 - \frac{\Delta t G_d}{2C_d}}{1 + \frac{\Delta t G_d}{2C_d}} v_d^{n-1} + \frac{\Delta t}{C_d} \frac{\{1 - S_{d+1}^{-1}\} i_{d+2}^{n-\frac{1}{2}} - \{1 - S_{d+2}^{-1}\} i_{d+1}^{n-\frac{1}{2}} + i_{\text{src},d}^{n-\frac{1}{2}}}{1 + \frac{\Delta t G_d}{2C_d}}, \quad (23)$$

$$i_d^{n+\frac{1}{2}} = \frac{1 - \frac{\Delta t R_d}{2L_d}}{1 + \frac{\Delta t R_d}{2L_d}} i_d^{n-\frac{1}{2}} - \frac{\Delta t}{L_d} \frac{\{S_{d+1} - 1\} v_{d+2}^n - \{S_{d+2} - 1\} v_{d+1}^n}{1 + \frac{\Delta t R_d}{2L_d}}. \quad (24)$$

2.3 Extended Method for General Dispersive Materials

In Sect. 2.2 the basic EC FDTD scheme for *non-dispersive* materials is derived. This has led to the emergence of the local impedance $Z(\omega) = j\omega L \propto j\omega\mu$ constituted of the inductance L (with resistance R for loss) and local admittance $Y(\omega) = j\omega C \propto j\omega\epsilon$ constituted of a capacitance C (with conductance G for loss), which are characteristic of non-dispersive media in the transmission line framework and correspond to the material non-dispersive constitutive parameters.

However, many problems in electrodynamics involve dispersive media, including *metamaterials* and *various natural materials*, for which application examples are presented in Sects. 5.1 and 5.2, respectively. In order to handle such problems, we generalize now the EC FDTD scheme to an *extended* EC FDTD scheme for dispersive materials. This extended scheme fundamentally consists in adding to the equivalent impedance $Z(\omega)$ and admittance $Y(\omega)$ the appropriate lumped elements to account for the specific dispersions considered, $\tilde{\mu}(\omega) = Z(\omega)/(j\omega\Delta_m)$ and $\tilde{\epsilon}(\omega) = Y(\omega)/(j\omega\Delta_e)$ [the lengths Δ_m and Δ_e are specified later]. This transformation will require additional update equations. The present section specifically focuses on *double-Drude* dispersion, which is the dispersion encountered in the so-called composite right/left-handed (CRLH) metamaterials [4]. Dispersions of the Lorentz and Debye types are also discussed.

The relation between the generalized-dispersive admittance and the corresponding dispersive function $\tilde{\epsilon}(\omega) = \epsilon_\infty + \Delta\epsilon'(\omega) - j\sigma_e/\omega - j\epsilon''(\omega)$ [where $\epsilon_\infty \doteq \tilde{\epsilon}(\omega \rightarrow \infty)$] reads

$$\tilde{\epsilon}_{d,\text{eff}}(\omega) = \frac{Y_d(\omega)}{j\omega} \frac{\Delta_d}{\frac{1}{4} \{1 + S_{d+1}^{-1} + S_{d+2}^{-1} + S_{d+1}^{-1} S_{d+2}^{-1}\} [\Delta_{d+1} \Delta_{d+2}]} = \frac{Y_d(\omega)}{j\omega \Delta_{e,d}}, \quad (25)$$

where $\Delta_{e,d}$ represents the formfactor of an artificial subcell with cross section $\bar{A}_d$ of the dual mesh and the height Δ_d of the main mesh and has the dimension of a length. Similarly, the dispersive function $\tilde{\mu}(\omega) = \mu_\infty + \Delta\mu'(\omega) - j\sigma_m/\omega - j\mu''(\omega)$ [where $\mu_\infty \doteq \tilde{\mu}(\omega \rightarrow \infty)$] can be determined from its corresponding generalized-dispersive impedance via

$$\tilde{\mu}_{d,\text{eff}}(\omega) = \frac{Z_d(\omega)}{j\omega} \frac{\frac{1}{2} \{1 + S_d^{-1}\} \Delta_d}{\Delta_{d+1} \Delta_{d+2}} = \frac{Z_d(\omega)}{j\omega \Delta_{m,d}}, \quad (26)$$

where $\Delta_{m,d}$ represents the formfactor of the artificial cell with cross section A_d of the main mesh subcell and height $\bar{\Delta}_d$ of the dual mesh subcell.

2.3.1 Drude Dispersion

Let us now particularize these generalized dispersive relations to the important case of *Drude* materials, which include for instance noble metals at optical frequencies (ϵ -Drude) or CRLH metamaterials (both ϵ and μ -Drude). In this case, the relevant admittance Y_d includes an inductance $L_{L,d}$ in parallel with the capacitance C_d as represented in Fig. 5(a) and the relevant impedance Z_d includes a capacitance $C_{L,d}$ in series with the inductance L_d , [4], as represented in Fig. 5(b). In terms of losses, we consider the most general situation, i.e., in series to each inductance there is a resistance, and parallel each capacitor a conductance considers the losses. The general dispersive parameters in Eqs. (25) and (26) become then

$$\begin{aligned} \tilde{\epsilon}_{\text{ext-Drude}}(\omega) &= \frac{Y_{\text{ext-Drude}}(\omega)}{j\omega \Delta_e} = \frac{C}{\Delta_e} \left\{ 1 - \frac{\frac{1}{L_L C}}{\omega \left(\omega - j \frac{R_L}{L_L} \right)} \right\} - j \frac{G}{\omega \Delta_e} \\ &= \epsilon_0 \epsilon_{r,\infty} \left\{ 1 - \frac{\omega_{pe}^2}{\omega (\omega - j\Omega)} \right\} - j \frac{\sigma_e}{\omega}, \end{aligned} \quad (27)$$

$$\begin{aligned} \tilde{\mu}_{\text{ext-Drude}}(\omega) &= \frac{Z_{\text{ext-Drude}}(\omega)}{j\omega \Delta_m} = \frac{L}{\Delta_m} \left\{ 1 - \frac{\frac{1}{L C_L}}{\omega \left(\omega - j \frac{G_L}{C_L} \right)} \right\} - j \frac{R}{\omega \Delta_m} \\ &= \mu_0 \mu_{r,\infty} \left\{ 1 - \frac{\omega_{pm}^2}{\omega (\omega - j\Omega)} \right\} - j \frac{\sigma_m}{\omega}. \end{aligned} \quad (28)$$

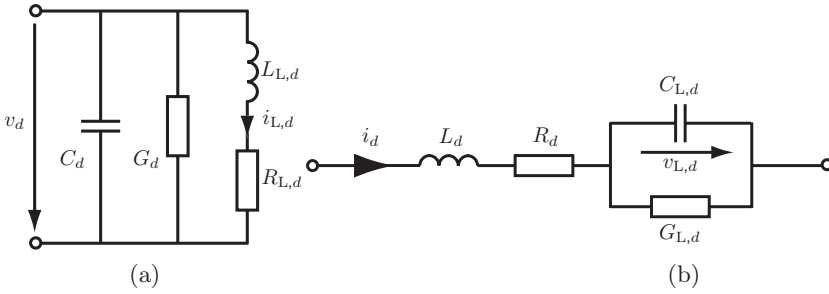


Fig. 5 Extended equivalent circuit models for Drude dispersive and lossy media. (a) Admittance $Y_{\text{ext-Drude}}(\omega)$ located at E -field nodes and related to the dispersive $\tilde{\epsilon}_{\text{ext-Drude}}(\omega)$. (b) Impedance $Z_{\text{ext-Drude}}(\omega)$ located at H -field nodes and related to the dispersive $\tilde{\mu}_{\text{ext-Drude}}(\omega)$

Due to the presence of the additional Drude lumped elements, the time-continuous Ampere's and Faraday's circuit law system of Eqs. (19) and (20) must be extended to the system

$$\{1 - S_{d+1}^{-1}\} i_{d+2} - \{1 - S_{d+2}^{-1}\} i_{d+1} + i_{\text{src},d} = +C_d \frac{d}{dt} v_d + G_d v_d + i_{L,d}, \quad (29)$$

$$v_d = +L_{L,d} \frac{d}{dt} i_{L,d} + R_{L,d} i_{L,d}, \quad (30)$$

$$\{S_{d+1} - 1\} v_{d+2} - \{S_{d+2} - 1\} v_{d+1} = -L_d \frac{d}{dt} i_d - R_d i_d - v_{L,d}, \quad (31)$$

$$i_d = +C_{L,d} \frac{d}{dt} v_{L,d} + G_{L,d} v_{L,d}, \quad (32)$$

where the two original equations include one more term and which has been increased by additional equations, as seen from the models of Fig. 5.

The final update equations representing the Drude counterparts of Eqs. (23) and (24) are finally obtained as

$$v_d^n = \frac{1 - \frac{\Delta t G_d}{2C_d}}{1 + \frac{\Delta t G_d}{2C_d}} v_d^{n-1} + \frac{\Delta t}{C_d} \frac{\{1 - S_{d+1}^{-1}\} i_{d+2}^{n-\frac{1}{2}} - \{1 - S_{d+2}^{-1}\} i_{d+1}^{n-\frac{1}{2}} - i_{L,d}^{n-\frac{1}{2}} + i_{\text{src},d}^{n-\frac{1}{2}}}{1 + \frac{\Delta t G_d}{2C_d}}, \quad (33)$$

$$i_{L,d}^{n+\frac{1}{2}} = \frac{1 - \frac{\Delta t R_{L,d}}{2L_{L,d}}}{1 + \frac{\Delta t R_{L,d}}{2L_{L,d}}} i_{L,d}^{n-\frac{1}{2}} + \frac{\Delta t}{L_{L,d}} \frac{v_d^n}{1 + \frac{\Delta t R_{L,d}}{2L_{L,d}}}, \quad (34)$$

$$i_d^{n+\frac{1}{2}} = \frac{1 - \frac{\Delta t R_d}{2L_d}}{1 + \frac{\Delta t R_d}{2L_d}} i_d^{n-\frac{1}{2}} - \frac{\Delta t}{L_d} \frac{\{S_{d+1} - 1\} v_{d+2}^n - \{S_{d+2} - 1\} v_{d+1}^n + v_{L,d}^n}{1 + \frac{\Delta t R_d}{2L_d}}, \quad (35)$$

$$v_{L,d}^n = \frac{1 - \frac{\Delta t G_{L,d}}{2C_{L,d}}}{1 + \frac{\Delta t G_{L,d}}{2C_{L,d}}} v_{L,d}^{n-1} + \frac{\Delta t}{C_{L,d}} \frac{i_d^{n-\frac{1}{2}}}{1 + \frac{\Delta t G_{L,d}}{2C_{L,d}}}. \quad (36)$$

2.3.2 Other Dispersive Media: Lorentz and Debye

Lorentz-type (one pole and one zero in ϵ or μ) dispersive materials may be analyzed by similar extended EC FDTD models [6]. Such Lorentz media are often encountered in natural materials (e.g. ferrimagnetic and ferromagnetic materials) and in metamaterials (e.g. split ring resonator structures). The circuit model for this type of material is shown in Fig. 6(a), and the extended EC FDTD scheme equations may be obtained following a procedure formally analogous to that for the Drude case.

As a final important example of dispersive media which can be analyzed by the extended EC FDTD scheme, let us consider the *Debye*-type dispersive case [typically characterized by the parameters ($\epsilon_\infty, \sigma_e, \Delta\epsilon_n, \omega_n = 1/\tau_n$)], which are very common in biological systems. A Debye material typically exhibits a permittivity function of the form

$$\begin{aligned} \tilde{\epsilon}_{N\text{-Debye}}(\omega) &= \frac{Y_{N\text{-Debye}}(\omega)}{j\omega\Delta\epsilon_e} = \frac{1}{\Delta\epsilon_e} \left\{ C_\infty - j\frac{G_0}{\omega} + \sum_{n=1}^N \frac{\Delta C_n}{1 + j\omega\Delta C_n R_n} \right\} \\ &= \epsilon_\infty - j\frac{\sigma_e}{\omega} + \sum_{n=1}^N \frac{\Delta\epsilon_n}{1 + j\omega/\omega_n}, \end{aligned} \quad (37)$$

which may include N poles and whose circuit model is shown in Fig. 6(b), while its permeability is usually constant and equal to μ_0 (non-magnetic medium). Again, the extended EC FDTD scheme equations may be obtained following a procedure formally analogous to that for the Drude case.

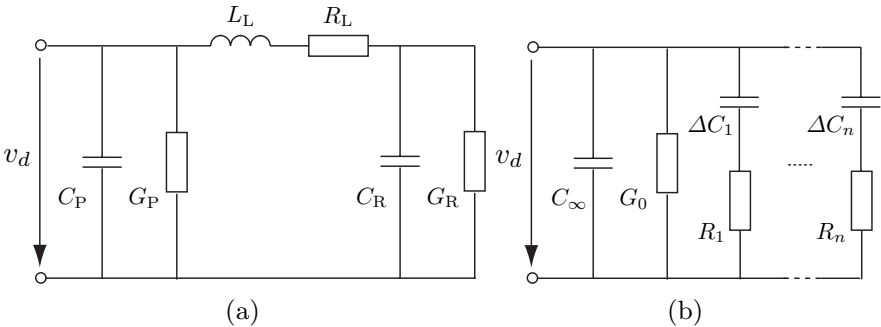


Fig. 6 Extended equivalent circuit admittance models for a Lorentz and a Debye dispersive and lossy media. (a) Extended lossy Lorentz model. (b) N -pole Debye circuit

3 Poynting Theorem for EC FDTD

This section presents the Poynting theorem for the EC FDTD scheme. It will show that the space-discrete Poynting theorem has exactly the same form as the space-continuous Poynting theorem in the time-continuous case but a slightly different form (with additional terms) in the time-discrete case. These results will be utilized in Sect. 4 to derive stability criteria. The procedure for the derivation of the space-discrete Poynting theorem is somewhat analogous to that for the space-continuous case. For generality, the derivations will be made for the double-Drude case. They can be easily particularized to the case of non-dispersive media or applied to other types of dispersive media.

3.1 Time-Continuous Case

In order to evaluate a Poynting quantity ($\mathbf{E} \times \mathbf{H}$, or scalar $E_d H_{d+1}$), one needs to consider perpendicular components of the E and H fields. Since the (Drude) current Eqs. (31) and (32) correspond to a H -field sample which is *parallel* to the E -field sample corresponding to the associated voltage Eqs. (29) and (30) (see Fig. 1, with $H_d \parallel E_d$), it is then necessary to consider, for one of the two fields, a different and perpendicular (i.e. *rotated*) component. Let us for instance keep the E_d sample unchanged and right-handedly *rotate* the H_d sample to select the appropriate perpendicular component, where the current and voltages considered are those orthogonal to each other; this is achieved by rotating or index-shifting ($d \rightarrow d+1$) Eqs. (31) and (32) by the operator R (Sect. 2.1).

The first step of the procedure consists then in multiplying Eqs. (29), (30), index-shifted (31) and index-shifted (32) with $v_d, i_{L,d}, (-i_{d+1})$ and $v_{L,d+1}$, respectively, and summing for each index-direction $d \in \{0, 1, 2\}$ over all the cells ($\mathbf{X} \in \Omega$) of the computational domain to determine its total power. This yields

$$\begin{aligned}
 & \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 \left(v_d \{1 - S_{d+1}^{-1}\} i_{d+2} - v_d \{1 - S_{d+2}^{-1}\} i_{d+1} + v_d i_{L,d} \right. \\
 & \quad \left. - i_{d+1} \{S_{d+2} - 1\} v_d + i_{d+1} \{S_d - 1\} v_{d+2} + v_{L,d+1} i_{d+1} \right) \\
 &= \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 \left(C_d v_d \frac{d}{dt} v_d + G_d v_d^2 + v_d i_{L,d} - v_d i_{\text{src},d} + L_{L,d} i_{L,d} \frac{d}{dt} i_{L,d} + R_{L,d} i_{L,d}^2 + \right. \\
 & \quad \left. L_{d+1} i_{d+1} \frac{d}{dt} i_{d+1} + R_{d+1} i_{d+1}^2 + v_{L,d+1} i_{d+1} + C_{L,d+1} v_{L,d+1} \frac{d}{dt} v_{L,d+1} + G_{L,d+1} v_{L,d+1}^2 \right). \tag{38}
 \end{aligned}$$

The different terms of this relation may be rearranged as

$$\begin{aligned}
\sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 P_{\text{src},d} &= \sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 v_d i_{\text{src},d} \\
&= \sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 \left(G_d v_d^2 + R_{L,d} i_{L,d}^2 + R_{d+1} i_{d+1}^2 + G_{L,d+1} v_{L,d+1}^2 \right. \\
&\quad \left. + \frac{d}{dt} \left[\frac{1}{2} C_d v_d^2 + \frac{1}{2} L_{L,d} i_{L,d}^2 + \frac{1}{2} L_{d+1} i_{d+1}^2 + \frac{1}{2} C_{L,d+1} v_{L,d+1}^2 \right] \right. \\
&\quad \left. - v_d i_{d+2} + v_d S_{d+1}^{-1} i_{d+2} + v_d i_{d+1} - v_d S_{d+2}^{-1} i_{d+1} \right. \\
&\quad \left. + S_{d+2} v_d i_{d+1} - v_d i_{d+1} - S_d v_{d+2} i_{d+1} + R^{-1} [v_d i_{d+2}] \right) \quad (39)
\end{aligned}$$

where the following different *total powers of the 3D EC mesh within the computational domain* are identified: (i) left-hand side: *power delivered by the sources* ($\int_{\Omega} \mathbf{E} \cdot \mathbf{J}_{\text{src}} \, dv$ in the field-continuous case); (ii) right-hand side – first row: *power dissipated*; (iii) right-hand side – second row: *power deriving from the net reactive energy stored* in the inductances and capacitances ($\frac{d}{dt} [W_m + W_e]$); right-hand side - third and fourth row: *power radiated within the volume* ($\int_{\Omega} \text{div}(\mathbf{E} \times \mathbf{H}) \, dv$ in the field-continuous case), where 4 out of the 8 products involving the terms $v_d i_{d+1}$ and $v_d i_{d+2}$ cancel out⁴.

Furthermore, applying spatial shifting (operator S and its inverse S^{-1}), it is found that the radiated power terms *inside the volume* Ω cancel out so that only the terms *at the surface* $\partial\Omega$ of this volume contribute to the radiated power, leading to the substitution: $\sum_{\mathbf{x} \in \Omega} \rightarrow \sum_{\mathbf{x} \in \partial\Omega}$. This fact constitutes the *discretized version of Gauss theorem* ($\int_{\Omega} \text{div}(\mathbf{E} \times \mathbf{H}) \, dv = \int_{\partial\Omega} (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{A}$). The previous relation may then be written in the form

$$\begin{aligned}
\sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 P_{\text{src},d} &= \sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 \left(P_{G_d} + P_{R_{L,d}} + P_{R_{d+1}} + P_{G_{L,d+1}} \right. \\
&\quad \left. + \frac{d}{dt} [W_{C_d} + W_{L_{L,d}} + W_{L_{d+1}} + W_{C_{L,d+1}}] \right) \\
&\quad + \sum_{d=0}^2 \left\{ \sum_{X_{d+1}} \sum_{X_{d+2}} \left(S_d v_{d+1} i_{d+2} \Big|_{X_d=N_{+,d}} - S_d v_{d+2} i_{d+1} \Big|_{X_d=N_{+,d}} \right. \right. \\
&\quad \left. \left. + v_{d+2} S_d^{-1} i_{d+1} \Big|_{X_d=N_{-,d}} - v_{d+1} S_d^{-1} i_{d+2} \Big|_{X_d=N_{-,d}} \right) \right\}, \quad (40)
\end{aligned}$$

which is the *EC FDTD discretized time-continuous version of the Poynting theorem*, stating that the power delivered by the sources is equal to the sum of the power

⁴ To indicate the cross-cancellation of the 1st and 8th term, the 8th has been index-shifted by the operator R^{-1} .

transmitted through the surface⁵ $\partial\Omega$, the power lost to heat in the volume Ω and the net reactive power stored in the volume Ω .

3.2 Time Discrete Case

Let us now derive the *time-discrete* counterpart of EC FDTD discretized Poynting theorem. This will lead to considerations of stability in the next section.

Multiplying the time-discretized versions of Eqs. (29), (30), (31) and (32) with $v_d^{n-\frac{1}{2}}$, $i_{L,d}^n$, $(-i_{d+1}^n)$ and $v_{L,d+1}^{n-\frac{1}{2}}$, respectively, yields

$$\begin{aligned}
 & \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 \left(v_d^{n-\frac{1}{2}} \{1 - S_{d+1}^{-1}\} i_{d+2}^{n-\frac{1}{2}} - v_d^{n-\frac{1}{2}} \{1 - S_{d+2}^{-1}\} i_{d+1}^{n-\frac{1}{2}} + v_d^n i_{L,d}^n \right. \\
 & \quad \left. - i_{d+1}^n \{S_{d+2} - 1\} v_d^n + i_{d+1}^n \{S_d - 1\} v_{d+2}^n + v_{L,d+1}^{n-\frac{1}{2}} i_{d+1}^{n-\frac{1}{2}} \right) \\
 & = \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 \left(C_d v_d^{n-\frac{1}{2}} \frac{v_d^n - v_d^{n-1}}{\Delta t} + G_d v_d^{n-\frac{1}{2}} \frac{v_d^n + v_d^{n-1}}{2} + v_d^{n-\frac{1}{2}} i_{L,d}^{n-\frac{1}{2}} - v_d^{n-\frac{1}{2}} i_{\text{src},d}^{n-\frac{1}{2}} \right. \\
 & \quad + L_{L,d} i_{L,d}^n \frac{i_{L,d}^{n+\frac{1}{2}} - i_{L,d}^{n-\frac{1}{2}}}{\Delta t} + R_{L,d} i_{L,d}^n \frac{i_{L,d}^{n+\frac{1}{2}} + i_{L,d}^{n-\frac{1}{2}}}{2} \\
 & \quad + L_{d+1} i_{d+1}^n \frac{i_{d+1}^{n+\frac{1}{2}} - i_{d+1}^{n-\frac{1}{2}}}{\Delta t} + R_{d+1} i_{d+1}^n \frac{i_{d+1}^{n+\frac{1}{2}} + i_{d+1}^{n-\frac{1}{2}}}{2} + v_{L,d+1}^n i_{d+1}^n \\
 & \quad \left. + C_{L,d+1} v_{L,d+1}^{n-\frac{1}{2}} \frac{v_{L,d+1}^n - v_{L,d+1}^{n-1}}{\Delta t} + G_{L,d+1} v_{L,d+1}^{n-\frac{1}{2}} \frac{v_{L,d+1}^n + v_{L,d+1}^{n-1}}{2} \right). \quad (41)
 \end{aligned}$$

Next partly replacing the time samples $v_d^{n-\frac{1}{2}}$, $i_{L,d}^n$, i_{d+1}^n and $v_{L,d+1}^{n-\frac{1}{2}}$ by their averages $\frac{1}{2}(v_d^n + v_d^{n-1})$, $\frac{1}{2}(i_{L,d}^{n+\frac{1}{2}} + i_{L,d}^{n-\frac{1}{2}})$, $\frac{1}{2}(i_{d+1}^{n+\frac{1}{2}} + i_{d+1}^{n-\frac{1}{2}})$ and $\frac{1}{2}(v_{L,d+1}^n + v_{L,d+1}^{n-1})$, respectively, transforms Eq. (41) into the re-arranged relation

$$\begin{aligned}
 \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 P_{\text{src},d}^{n-\frac{1}{2}} & = \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 v_d^{n-\frac{1}{2}} i_{\text{src},d}^{n-\frac{1}{2}} = \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 \frac{1}{2} (v_d^n + v_d^{n-1}) i_{\text{src},d}^{n-\frac{1}{2}} \\
 & = \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 \left(G_d \left(v_d^{n-\frac{1}{2}} \right)^2 + R_{L,d} \left(i_{L,d}^n \right)^2 + R_{d+1} \left(i_{d+1}^n \right)^2 + G_{L,d+1} \left(v_{L,d+1}^{n-\frac{1}{2}} \right)^2 \right. \\
 & \quad \left. + \frac{1}{2} C_d \frac{(v_d^n)^2 - (v_d^{n-1})^2}{\Delta t} + \frac{1}{2} L_{L,d} \frac{\left(i_{L,d}^{n+\frac{1}{2}} \right)^2 - \left(i_{L,d}^{n-\frac{1}{2}} \right)^2}{\Delta t} \right)
 \end{aligned}$$

⁵ The indices values $N_{-,d}$ and $N_{+,d}$ in the last summation of Eq. (40) are the lower and upper bounds of the cuboidal domain Ω in the direction d . The inner two sums over X_{d+1} and X_{d+2} are due to the summation over the plane boundaries of the volume Ω .

$$\begin{aligned}
& + \frac{1}{2} L_{d+1} \frac{\left(i_{d+1}^{n+\frac{1}{2}}\right)^2 - \left(i_{d+1}^{n-\frac{1}{2}}\right)^2}{\Delta t} + \frac{1}{2} C_{L,d+1} \frac{\left(v_{L,d+1}^n\right)^2 - \left(v_{L,d+1}^{n-1}\right)^2}{\Delta t} \\
& + \frac{1}{2} \left[-v_d^n i_{d+2}^{n-\frac{1}{2}} - v_d^{n-1} i_{d+2}^{n-\frac{1}{2}} + v_d^n S_{d+1}^{-1} i_{d+2}^{n-\frac{1}{2}} + v_d^{n-1} S_{d+1}^{-1} i_{d+2}^{n-\frac{1}{2}} \right. \\
& + v_d^n i_{d+1}^{n-\frac{1}{2}} + v_d^{n-1} i_{d+1}^{n-\frac{1}{2}} - v_d^n S_{d+2}^{-1} i_{d+1}^{n-\frac{1}{2}} - v_d^{n-1} S_{d+2}^{-1} i_{d+1}^{n-\frac{1}{2}} \\
& + S_{d+2} v_d^n i_{d+1}^{n+\frac{1}{2}} + S_{d+2} v_d^n i_{d+1}^{n-\frac{1}{2}} - v_d^n i_{d+1}^{n+\frac{1}{2}} - v_d^n i_{d+1}^{n-\frac{1}{2}} \\
& - S_d v_{d+2}^n i_{d+1}^{n+\frac{1}{2}} - S_d v_{d+2}^n i_{d+1}^{n-\frac{1}{2}} + v_{d+2}^n i_{d+1}^{n+\frac{1}{2}} + R^{-1} \left[v_d^n i_{d+2}^{n-\frac{1}{2}} \right] \\
& - v_d^n i_{L,d}^{n+\frac{1}{2}} - v_d^n i_{L,d}^{n-\frac{1}{2}} - v_{L,d+1}^n i_{d+1}^{n-\frac{1}{2}} - v_{L,d+1}^{n-1} i_{d+1}^{n-\frac{1}{2}} \\
& \left. + v_d^n i_{L,d}^{n-\frac{1}{2}} + v_d^{n-1} i_{L,d}^{n-\frac{1}{2}} + v_{L,d+1}^n i_{d+1}^{n+\frac{1}{2}} + v_{L,d+1}^n i_{d+1}^{n-\frac{1}{2}} \right], \quad (42)
\end{aligned}$$

which includes 16 non-dispersive and 8 dispersive vi products, over which 4 non-dispersive and 4 dispersive cancel out⁶. As for the time-continuous case, applying spatial shifting (operator S and its inverse S^{-1}), it is found that some of the radiated power terms *inside the volume* Ω cancel out and only 4 surface terms remain. The previous relation may then be written in the form

$$\begin{aligned}
\sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 P_{\text{src},d}^{n-\frac{1}{2}} &= \sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 \left(P_{G_d}^{n-\frac{1}{2}} + P_{R_{L,d}}^n + P_{R_{d+1}}^n + P_{G_{L,d+1}}^{n-\frac{1}{2}} + \frac{W_{C_d}^n - W_{C_d}^{n-1}}{\Delta t} \right. \\
& + \frac{W_{L_{L,d}}^{n+\frac{1}{2}} - W_{L_{L,d}}^{n-\frac{1}{2}}}{\Delta t} + \frac{W_{L_{d+1}}^{n+\frac{1}{2}} - W_{L_{d+1}}^{n-\frac{1}{2}}}{\Delta t} + \frac{W_{C_{L,d+1}}^n - W_{C_{L,d+1}}^{n-1}}{\Delta t} \\
& + \frac{1}{2} \left[-v_d^n \{1 - S_{d+2}^{-1}\} i_{d+1}^{n+\frac{1}{2}} + v_d^{n-1} \{1 - S_{d+2}^{-1}\} i_{d+1}^{n-\frac{1}{2}} \right. \\
& + v_d^n \{1 - S_{d+1}^{-1}\} i_{d+2}^{n+\frac{1}{2}} - v_d^{n-1} \{1 - S_{d+1}^{-1}\} i_{d+2}^{n-\frac{1}{2}} \\
& \left. - v_d^n i_{L,d}^{n+\frac{1}{2}} + v_d^{n-1} i_{L,d}^{n-\frac{1}{2}} + v_{L,d+1}^n i_{d+1}^{n+\frac{1}{2}} - v_{L,d+1}^{n-1} i_{d+1}^{n-\frac{1}{2}} \right] \Bigg) \\
& + \sum_{d=0}^2 \left\{ \sum_{X_{d+1}} \sum_{X_{d+2}} \left(S_d v_{d+1}^n i_{d+2}^n |_{X_d=N_{+,d}} - S_d v_{d+2}^n i_{d+1}^n |_{X_d=N_{+,d}} \right. \right. \\
& \left. \left. + v_{d+2}^{n-\frac{1}{2}} S_d^{-1} i_{d+1}^{n-\frac{1}{2}} |_{X_d=N_{-,d}} - v_{d+1}^{n-\frac{1}{2}} S_d^{-1} i_{d+2}^{n-\frac{1}{2}} |_{X_d=N_{-,d}} \right) \right\}, \quad (43)
\end{aligned}$$

⁶ To indicate the cross-cancellation of the 1st and 16th term, the 16th has been index-shifted by the operator R^{-1} .

which is the time-discrete EC FDTD version of the Poynting theorem⁷. In contrast to its time-continuous counterpart [Eq. (40)], this relation does not simplify to a form exactly identical to the space-continuous Poynting theorem: 12 *vi* “time-discretization-compensatory” product terms (8 for the non-dispersive case and 4 additional ones for the double Drude material), due to the two different time steps $n - \frac{1}{2}$ and n involved, remain present in the power expressions.

4 Stability Criteria Based on a Lyapunov Energy

Time-discretized Maxwell’s equations are subject to instability when the time step Δt is larger than the computational cell size Δ divided by the speed of light c , and multiplied by a factor depending on the dimension of the computational domain. Whereas in the case of *non-dispersive* media stability is ensured by following the simple Courant–Friedrichs–Lewy criterion [$\Delta t < \Delta/(\sqrt{3}c)$] [1], no explicit guaranteed-stability criterion is available for *dispersive* media in FDTD schemes. The EC FDTD scheme provides such a criterium, which will be derived now applying a Lyapunov energy approach to the time-discrete Poynting theorem [Eq. (43)]. Here the volume Ω in the discrete Poynting theorem must be the whole computational domain in order to include the overall energy in the 3D EC network.

In order to simplify the derivation, we assume that the computational domain is truncated either by PECs, by PMCs or by lossy material for absorption (with PEC or PMC walls behind)⁸, so that the last sum in Eq. (43) vanishes.

The *Lyapunov energy stability criterium* states that a system is stable if its energy: (1) is a positive definite scalar function, and (2) is constant or monotonically decreasing after all sources have been switched off [7].

Defining as a generalized energy expression the Lyapunov function

$$L_{\text{global}}^n = \sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 \left(W_{C_d}^n + W_{L,d}^{n+\frac{1}{2}} + W_{L,d+1}^{n+\frac{1}{2}} + W_{C_{L,d+1}}^n \right. \\ \left. + \frac{\Delta t}{2} \left[-v_d^n \{1 - S_{d+2}^{-1}\} i_{d+1}^{n+\frac{1}{2}} + v_d^n \{1 - S_{d+1}^{-1}\} i_{d+2}^{n+\frac{1}{2}} - v_d^n i_{L,d}^{n+\frac{1}{2}} + v_{L,d+1}^n i_{d+1}^{n+\frac{1}{2}} \right] \right), \quad (44)$$

the time-discrete Poynting theorem [Eq. (43)] may be reformulated as

$$\frac{L_{\text{global}}^n - L_{\text{global}}^{n-1}}{\Delta t} = \sum_{\mathbf{x} \in \Omega} \sum_{d=0}^2 \left(P_{\text{src},d}^{n-\frac{1}{2}} - P_{G_d}^{n-\frac{1}{2}} - P_{R_{L,d}}^n - P_{R_{d+1}}^n - P_{G_{L,d+1}}^{n-\frac{1}{2}} \right) \leq 0 \\ \text{for } P_{\text{src},d}^{n-\frac{1}{2}} = 0, \quad (45)$$

⁷ The indices values $N_{-,d}$ and $N_{+,d}$ in the last summation of Eq. (43) are the lower and upper bounds of the cuboidal domain Ω in the direction d . The inner two sums over X_{d+1} and X_{d+2} are due to the summation over the plane boundaries of the volume Ω .

⁸ Special absorbing boundary conditions of traveling-wave or perfectly matched layer type are not considered.

which means that the global energy of the system remains constant or decreases in time after the source has been switched off, and the second Lyapunov condition is thus fulfilled.

The demonstration of the first Lyapunov condition, namely the positive definiteness of the Lyapunov function, or equivalently the properness of L_{global}^n in Eq. (44) [7], is more intricate. The sought stability criterion will be obtained in terms of the maximum Δt delimiting the range where L_{global}^n is positive-definite.

Since the positive-definiteness of the global energy function L_{global}^n is difficult to establish, we decompose it in terms of *the local Lyapunov function* $L_{\text{local},d}^n(\mathbf{X})$

$$L_{\text{global}}^n = \sum_{\mathbf{X} \in \Omega} \sum_{d=0}^2 L_{\text{local},d}^n, \quad (46)$$

which involves only local voltages and currents. Once the positive definiteness of this local function will have been shown, the positive definiteness of the global one will follow by superposition.

For the non-dispersive case each summand of the global Lyapunov energy function [Eq. (44)] includes the two orthogonal values v_d and i_{d+1} (via the energies W_{C_d} and $W_{L_{d+1}}$) and additionally the adjacent currents⁹ i_{d+2} , and (the negatively shifted) $S_{d+2}^{-1}i_{d+1}$ and $S_{d+1}^{-1}i_{d+2}$. This local configuration is also used for the $L_{\text{local},d}^n$. But instead of just considering the energy of the the current i_{d+1} as in Eq. (44), the energies of all four surrounding currents are included. If this arrangement is summed up over all directions d and all $\mathbf{X}$ as defined in Eq. (46) the energies of the four surrounding currents would have been counted four times. Thus a weighting factor of 1/4 must be used for proper energy summation. The advantage of the mentioned re-arrangement of the energy terms is the fact that such a local Lyapunov function can be evaluated by a quadratic form. The positive definiteness of $L_{\text{local},d}^n$ is equivalent to the positive definiteness of the associated quadratic form matrix $\mathbf{A}$. This can be mostly done by applying Sylvester's criterion, which states that all of the leading principal minors of the square matrix must have a positive determinant for the matrix to be positive definite.

Within the following subsections three important practical cases are considered for stability analysis: the non-dispersive material, the plasmonic ϵ -Drude case and finally the double Drude metamaterial.

4.1 Stability Criterion for Non-Dispersive Materials

By using the advantageous computational configuration for $L_{\text{local},d}^n(\mathbf{X})$ where the admittance Y_d is at its center together with only one quarter (above-mentioned weighting factors) of the four adjacent and orthogonal impedances, the quadratic form

⁹ These terms are due to the remaining 4 vi products.

including only the non-dispersive reactive EC elements¹⁰ and the voltages/current vector Θ_5 reads¹¹

$$\begin{aligned}
 2 L_{\text{local},d}^n &= \Theta_5^T \cdot \mathbf{A}_5 \cdot \Theta_5 \\
 &= \Theta_5^T \cdot \begin{pmatrix} C_d & -\frac{\Delta t}{2} & +\frac{\Delta t}{2} & +\frac{\Delta t}{2} & -\frac{\Delta t}{2} \\ -\frac{\Delta t}{2} & \frac{L_{d+1}}{4} & 0 & 0 & 0 \\ +\frac{\Delta t}{2} & 0 & \frac{S_{d+2}^{-1} L_{d+1}}{4} & 0 & 0 \\ +\frac{\Delta t}{2} & 0 & 0 & \frac{L_{d+2}}{4} & 0 \\ -\frac{\Delta t}{2} & 0 & 0 & 0 & \frac{S_{d+1}^{-1} L_{d+2}}{4} \end{pmatrix} \cdot \begin{pmatrix} v_d^n \\ i_{d+1}^{n+\frac{1}{2}} \\ S_{d+2}^{-1} i_{d+1}^{n+\frac{1}{2}} \\ i_{d+2}^{n+\frac{1}{2}} \\ S_{d+1}^{-1} i_{d+2}^{n+\frac{1}{2}} \end{pmatrix} \quad (47)
 \end{aligned}$$

Sylvester's criterion demands that all of the leading principal minors of the square matrix must have a positive determinant in order to have a positive overall matrix, which leads to several stability criteria. Since the determinant of the overall matrix $\mathbf{A}_5$ is the most limiting one for the time step, we have for the maximal $\Delta t_{\text{non-disp}}$

$$\Delta t_{\text{non-disp}} < \frac{2}{\max_{\mathbf{x} \in \Omega, d \in \{0,1,2\}} \sqrt{\frac{1}{C_d} \left(\frac{4}{L_{d+1}} + \frac{4}{S_{d+2}^{-1} L_{d+1}} + \frac{4}{L_{d+2}} + \frac{4}{S_{d+1}^{-1} L_{d+2}} \right)}}. \quad (48)$$

For an uniform mesh the relation $\Delta t_{\text{non-disp}} = \sqrt{3/4} \Delta t_{\text{CFL}} \approx 87\% \Delta t_{\text{CFL}}$ holds, where Δt_{CFL} is the conventional time step based on the Courant-Friedrich-Levy (CFL) condition [1]. However, in non-uniform (graded) meshes, where the smallest cell is adjacent to a larger one (with at least 33% longer edges), the EC-based criterion yields a larger time step. This improvement depends on the size-ratio of directly adjacent cells. For extreme meshing situations an improvement factor $\Delta t_{\text{non-disp}}/\Delta t_{\text{CFL}}$ of 10 can easily be achieved. The ordinary CFL criteria has been derived for an uniform mesh and a homogeneous material distribution, therefore it seems to be too conservative in highly graded meshes, since the smallest cell size has to be used in the formula [1].

4.2 Stability Criterion for Plasmonic (ϵ -Drude) Materials

For a plasmonic (ϵ -Drude) medium, which has been deployed in the optical coupler example to be presented in Sect. 5.2, the admittances at the edges of the EC unit cell [Fig. 4] are extended by a LH inductance L_L in parallel, and the quadratic form reads

¹⁰ All elements with an index L are set to zero.

¹¹ The index 5 should indicate the number of components of this vector.

$$\Theta_6^T \cdot \mathbf{A}_6 \cdot \Theta_6 = \Theta_6^T \cdot \begin{pmatrix} C_d & -\frac{\Delta t}{2} & -\frac{\Delta t}{2} & +\frac{\Delta t}{2} & +\frac{\Delta t}{2} & -\frac{\Delta t}{2} \\ -\frac{\Delta t}{2} & L_{L,d} & 0 & 0 & 0 & 0 \\ -\frac{\Delta t}{2} & 0 & \frac{L_{d+1}}{4} & 0 & 0 & 0 \\ +\frac{\Delta t}{2} & 0 & 0 & \frac{S_{d+2}^{-1} L_{d+1}}{4} & 0 & 0 \\ +\frac{\Delta t}{2} & 0 & 0 & 0 & \frac{L_{d+2}}{4} & 0 \\ -\frac{\Delta t}{2} & 0 & 0 & 0 & 0 & \frac{S_{d+1}^{-1} L_{d+2}}{4} \end{pmatrix} \cdot \begin{pmatrix} v_d^n \\ i_{L,d}^{n+\frac{1}{2}} \\ i_{d+1}^{n+\frac{1}{2}} \\ S_{d+2}^{-1} i_{d+1}^{n+\frac{1}{2}} \\ i_{d+2}^{n+\frac{1}{2}} \\ S_{d+1}^{-1} i_{d+2}^{n+\frac{1}{2}} \end{pmatrix}, \quad (49)$$

which yields by Sylvester's criterion (6×6 -matrix determinant) the maximal possible Δt for an ε -Drude material

$$\Delta t_{\varepsilon\text{-Drude}} < \frac{2}{\max_{\mathbf{x} \in \Omega, d \in \{0,1,2\}} \sqrt{\frac{1}{C_d} \left(\frac{1}{L_{L,d}} + \frac{4}{L_{d+1}} + \frac{4}{S_{d+2}^{-1} L_{d+1}} + \frac{4}{L_{d+2}} + \frac{4}{S_{d+1}^{-1} L_{d+2}} \right)}} \quad (50)$$

Inspection of Eq. (50) confirms that a smaller time step limit must be used compared to the non-dispersive case [$\Delta t_{\varepsilon\text{-Drude}} < \Delta t_{\text{non-disp}}$] due to the additional term $1/L_{L,d}$ in the denominator. This fact makes the novel criterion even more important for stable time domain simulation of plasmonic structures.

4.3 Stability Criterion for Double Drude Materials

For a double Drude (ε -Drude and μ -Drude) medium, the admittances at the edges of the EC unit cell are extended by a LH inductance L_L in parallel and the impedances in the faces of the EC unit cell [Fig. 4] are extended by a LH capacitance C_L in series. In this case the voltage/current vector reads

$$\Theta_{10}^T \doteq \left(v_d^n, i_{L,d}^{n+\frac{1}{2}}, i_{d+1}^{n+\frac{1}{2}}, S_{d+2}^{-1} i_{d+1}^{n+\frac{1}{2}}, i_{d+2}^{n+\frac{1}{2}}, S_{d+1}^{-1} i_{d+2}^{n+\frac{1}{2}}, v_{L,d+1}^n, S_{d+2}^{-1} v_{L,d+1}^n, v_{L,d+2}^n, S_{d+1}^{-1} v_{L,d+2}^n \right), \quad (51)$$

i.e. the equation $|\mathbf{A}_{10}|(\Delta t) > 0$ must be solved for Δt . In principle this might be possible, but the result will be very cumbersome. Therefore the special case

$$\frac{1}{\omega_{\text{se},d}^2} \doteq L_{d+1} C_{L,d+1} = S_{d+2}^{-1} L_{d+1} S_{d+2}^{-1} C_{L,d+1} = L_{d+2} C_{L,d+2} = S_{d+1}^{-1} L_{d+2} S_{d+1}^{-1} C_{L,d+2} \quad (52)$$

is considered here instead, due to a simplified determination of the stability criterion. Solving the simplified determinant equation yields the maximal possible Δt for a double-Drude material

$$\begin{aligned} \Delta t_{\text{double-Drude}} &< \frac{2}{\max_{\mathbf{X} \in \Omega, d \in \{0,1,2\}} \sqrt{\omega_{\text{sh},d}^2 + \omega_{\text{se},d}^2 + \omega_{\text{mix},d}^2}} \\ &< \frac{2}{\max_{\mathbf{X} \in \Omega, d \in \{0,1,2\}} \omega_{\text{double-Drude},d}} \end{aligned} \quad (53)$$

with

$$\begin{aligned} \omega_{\text{double-Drude},d}^2 &= \frac{1}{2} \left(\omega_{\text{sh},d}^2 + \omega_{\text{se},d}^2 + \omega_{\text{mix},d}^2 \right. \\ &\quad \left. + \sqrt{\omega_{\text{sh},d}^4 + \omega_{\text{se},d}^4 + \omega_{\text{mix},d}^4 + 2 \left(\omega_{\text{se},d}^2 \omega_{\text{mix},d}^2 + \omega_{\text{sh},d}^2 \omega_{\text{mix},d}^2 - \omega_{\text{sh},d}^2 \omega_{\text{se},d}^2 \right)} \right) \\ &< \omega_{\text{sh},d}^2 + \omega_{\text{se},d}^2 + \omega_{\text{mix},d}^2 \end{aligned} \quad (54)$$

where

$$\omega_{\text{sh},d}^2 = \frac{1}{L_{L,d} C_d}, \quad (55)$$

$$\omega_{\text{mix},d}^2 = 16 \omega_{\text{sh},d}^2 \omega_{\text{se},d}^2 L_d \bar{C}_{L,\text{se},d} \quad \text{with} \quad (56)$$

$$\bar{C}_{L,\text{se},d} = \frac{1}{4} (C_{L,d+1} + S_{d+2}^{-1} C_{L,d+1} + C_{L,d+2} + S_{d+1}^{-1} C_{L,d+2}). \quad (57)$$

The simplified denominator $[\omega_{\text{sh},d}^2 + \omega_{\text{se},d}^2 + \omega_{\text{mix},d}^2]$ in equation (53) yields a slightly lower limit for the time step, but is easier to compute. The term is an upper estimate for the complicated $\omega_{\text{double-Drude},d}$. Inspection of Eq. (53) reveals that a smaller time step limit must be used compared to the non-dispersive and also the ε -Drude case $[\Delta t_{\text{double-Drude}} < \Delta t_{\varepsilon\text{-Drude}} < \Delta t_{\text{non-disp}}]$, due to the additional term $\omega_{\text{se},d}^2$ in the denominator of Eq. (53).

5 Simulation Examples

This section presents examples of dispersive material structures and devices computed by the EC FDTD method. The dispersive materials used are *metamaterials* in Sect. 5.1 and *natural materials* in Sect. 5.2.

5.1 Applications based on Dispersive Metamaterials

A *metamaterial* is an artificial structure which is electromagnetically homogeneous (i.e. possesses a structural unit cell much smaller than the guided wavelength,

$p \ll \lambda_g$) and which exhibits unusual properties not readily found in nature, such as for instance a negative refractive index (NRI) [4]. Metamaterials are generally dispersive media. This section presents a spatial-frequency refractive filter and an open surface-plasmonic resonator. In addition, it discusses a few other possible metamaterial applications.

5.1.1 Spatial-Frequency Refractive Filter

The spatial-frequency refractive filter is shown in Fig. 7. It consists of a composite right/left-handed (CRLH) [4] wedge placed in vacuum. This wedge is excited by a two-tone Gaussian (plane) wave with the lower frequency (f_l) in its left-handed range and the higher frequency (f_h) in its right-handed range. As a consequence, the lower frequency part of the signal experiences negative refraction and is therefore refracted toward the left, while the higher frequency part of the signal experiences positive refraction and is therefore refracted toward the right. The structure presented is specifically designed such that, with a 11.3° inclined-plane wedge, the output beams refract under the angles $\pm 30^\circ$ with respect to the direction of the incident wave. The structure operates thus as a spatial ($\pm\theta$) – frequential (f_l/f_h) filter. Such a device may be used as quasi-optical diplexer.

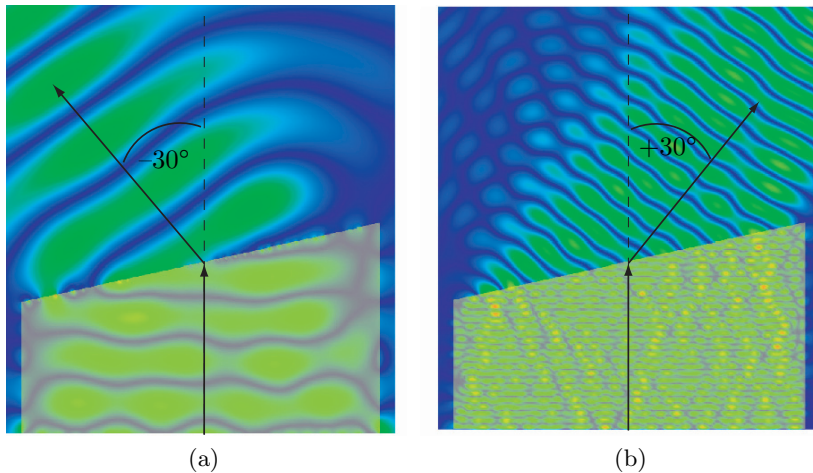


Fig. 7 Spatial-frequency refractive filter, consisting of a CRLH wedge excited by a two-tone signal. **(a)** Negatively refracted lower-frequency beam, for which $n(4.21 \text{ GHz}) = -1.63$. **(b)** Positively refracted higher frequency beam, for which $n(12.56 \text{ GHz}) = +3.37$

5.1.2 Open Surface-Plasmonic Resonator

The open surface-plasmonic resonator is shown in Fig. 8(a). It consists of four NRI metamaterial (MTM) triangular regions with $n = -1$ interleaved with comple-

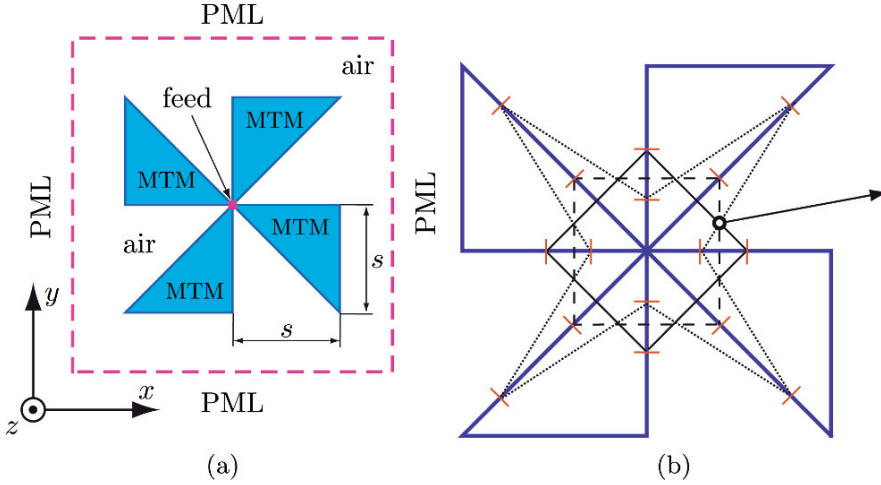


Fig. 8 Open surface plasmonic resonator. The metamaterial (MTM) regions are made of negative refractive index (NRI) medium with $n = -1$ while the complementary regions are vacuum ($n = +1$). The structure is bounded in the xy -plane by 4 perfect matched layers (PMLs) for absorption and in the z -direction by two perfect electric conductor (PEC) walls (bottom and top) for TM_z operation. (a) Top view of the structure. (b) Ray-tracing description, for a source located at the position of the black circle (3 rays at different angles are shown).

mentary vacuum ($n = +1$) regions, and this arrangement is bounded by perfectly matched layer (PML) absorbing boundaries on the sides and by perfect electric (PEC) walls at the bottom and at the top to ensure TM_z operation.

Figure 8(b) shows a naive ray-tracing description of the expected behavior of the resonator: due to the negative and iso-density refraction occurring at each interface, the electromagnetic wave radiated by a source within the structure should be “trapped”, as indicated, despite the fact that no walls surround the structure to confined the energy as in a conventional cavity resonator (hence the term “open” resonator). In addition, it is known that the evanescent components (corresponding to the near-field radiation contributions) of a wave incident at the interface between a right-handed and a NRI medium excite *surface plasmons* along the interface [8], as suggested in Fig. 9(a). The time snapshots of Figs. 9(b), (c), (d), (e) and (f) illustrate the refractive trapping and surface plasmonic behavior of the resonator. Note that only a small amount of energy is radiated through the open boundaries. This type of “energy traps” may lead to high-quality factor resonators, due to the absence of lossy metal walls, if purely dielectric NRI media are used.

5.1.3 Other Metamaterial Structures

Many other types of metamaterials structures may be easily modeled by the proposed EC FDTD method. CRLH metamaterials are double Drude (only one zero

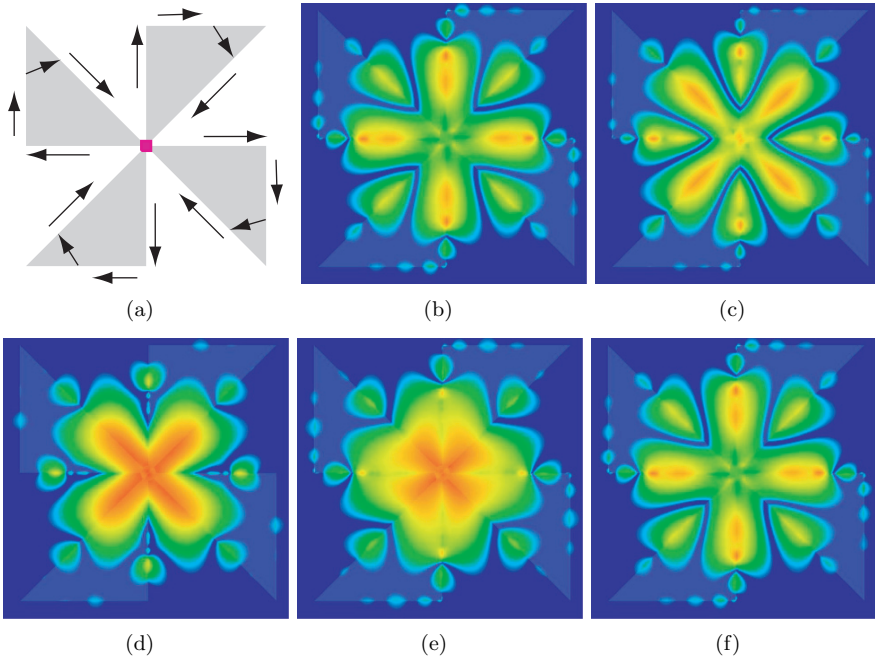


Fig. 9 Open surface-plasmonic resonator. (a) Surface plasmon propagation, and distribution of the vertical component of the electric field $[E_z(x,y)]$ at the resonance frequency f_{res} at the phase instants (b) 0° , (c) 45° , (d) 90° , (e) 135° , (f) 180° . Here, the excitation is located at the center of the structure

in both ϵ and μ) media $[\epsilon(\omega)p = C_R - 1/(\omega^2 L_L)$ and $\mu(\omega)p = L_R - 1/(\omega^2 C_L)]$. Other metamaterials are double Lorentz (one pole and one zero in both ϵ and μ) [6] or mixed Drude-Lorentz as the thin wire and split ring resonator structure which was the first experimentally verified NRI metamaterial [9] (Fig. 10), and more com-

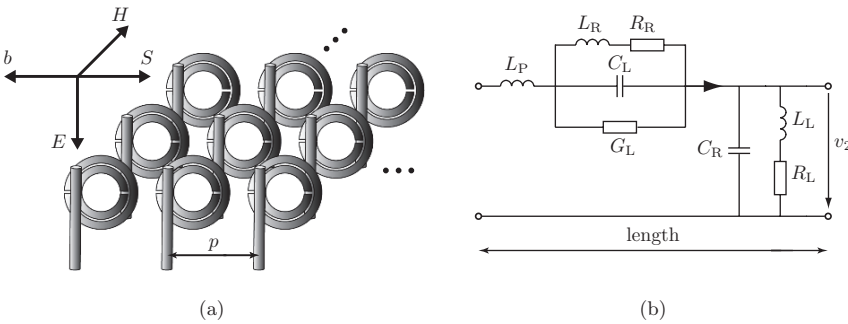


Fig. 10 Firstly reported practical left-handed metamaterial structure (a) of the mixed Drude-Lorentz short wire and split ring resonator kind [9] and its EC FDTD model (b). The series inductance L_P and shunt capacitance C_R represent the space wave propagating between the wire and ring scatterers, the impedance anti-tank $L_R - C_L$ models the split ring resonator (Lorentz) magnetic dipolar effect and the admittance inductance L_L models the wire (Drude) electric dipolar effect

plex possibilities exist. Corresponding dispersive EC FDTD models are straightforwardly available in the proposed EC FDTD scheme.

5.2 Applications Based on Natural Dispersive Materials

Many natural materials are dispersive. This includes for instance plasmas, ferromagnetic and ferroelectric materials, noble metals at optical frequencies and biological tissues. The EC FDTD method, by including various circuit-based dispersive (and also lossy) models both for ϵ and for μ (Drude, Lorentz, Debye, and variations thereof) can ideally analyze structures and devices made of such materials. This section presents the example of an optical coupled surface plasmon subwavelength power divider and discusses the applicability of the method to biological tissues.

5.2.1 Optical Coupled Surface Plasmon Subwavelength Power Divider

The optical power divider to be discussed here [10] is essentially constituted of a thin metallic Ag film within SiO_2 ($\epsilon_{\text{SiO}_2} = 2.09$) and supporting two surface (in fact interface) surface plasmons coupling to each other. It is appropriate to first provide an accurate description of the plasmonic response of Ag at optical frequencies.

It is well known that noble metals, such as silver (Ag), exhibit at optical frequencies a Drude-type permittivity response

$$\tilde{\epsilon}_{\text{Drude}}(\omega) = \epsilon_0 \left\{ 1 - \frac{\omega_p^2}{\omega(\omega - j\Omega)} \right\}. \quad (58)$$

This expression with the parameters $\omega_p = 12.9 \times 10^{15} \text{ rad/s}$, $\Omega = 8.0 \times 10^{13} \text{ rad/s}$ was used for instance in [11] for the design of a plasmonic contra-directional coupler. Since this simple model fits experimental data [12] only in a very narrow-band region around 500 THz, we propose the following extended model

$$\tilde{\epsilon}_{\text{ext-Drude}}(\omega) = \epsilon_0 \epsilon_{r,\infty} \left\{ 1 - \frac{\omega_p^2}{\omega(\omega - j\Omega)} \right\} - j \frac{\sigma_0}{\omega}, \quad (59)$$

which includes the two additional parameters $\epsilon_{r,\infty}$ (asymptotic term) and σ_0 (conductive term) to better fit experimental data in the frequency range of interest. The complex permittivity $\epsilon_r = \epsilon_r' - j\epsilon_r''$ obtained from this model for Ag with $\epsilon_{r,\infty} = 3.9$, $\omega_p = 7.0 \times 10^{15} \text{ rad/s}$, $\Omega = 2.3 \times 10^{13} \text{ rad/s}$ and $\sigma_0 = 8.0 \times 10^3 \text{ S/m}$, yielding the best fit to the measured values of [12] in the frequency range of 450 THz–850 THz, is shown in Fig. 11.

The optical coupled surface plasmon subwavelength power divider is presented in Fig. 12 along with its field distribution. It consists of a tapered metal (Ag) film embedded in a dielectric (SiO_2) material, with the coupling region in the center

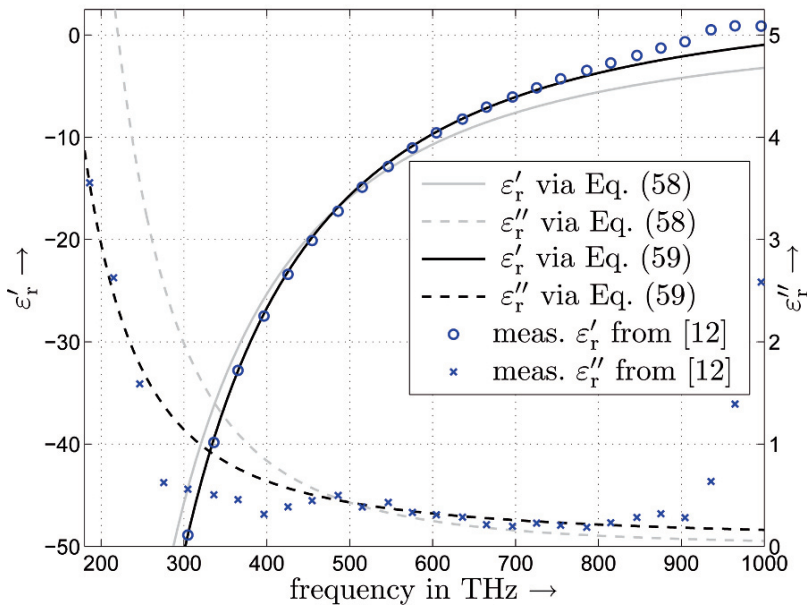


Fig. 11 Measured complex permittivity $\epsilon_r = \epsilon'_r - j\epsilon''_r$ from [12] compared to the data obtained by the simple Drude model given in Eq. (58) and an extended Drude function given in Eq. (59)

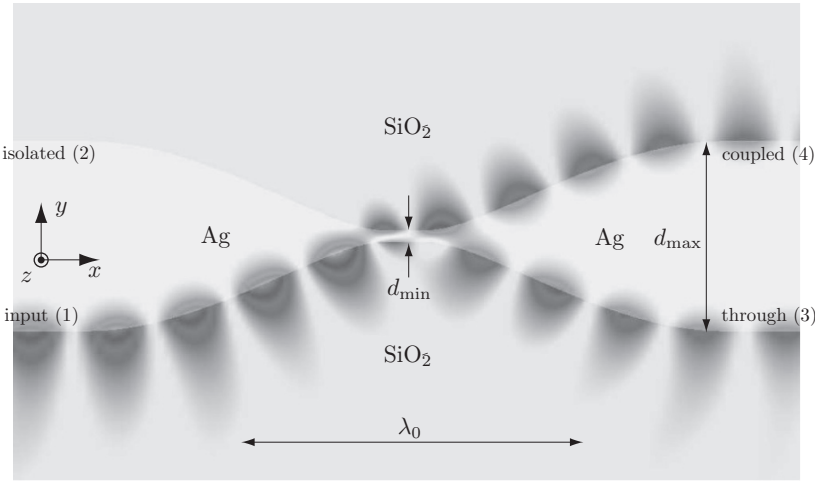


Fig. 12 Optical coupled surface plasmon subwavelength 3-dB/quadrature power divider (*longitudinal section view*) with intensity plot of the driven-mode TE_z magnetic field $H_z(x, y; f = 770\text{THz})$

and the input/output regions on either sides. The field distribution shown indicates 3-dB/quadrature splitting of the incoming guided surface plasmon. The dimensions are $d_{\min} = 15$ nm, $d_{\max} = 240$ nm, $\ell_{\text{narrow}} = 50$ nm and $\ell_{\text{taper}} = 375$ nm yielding an overall length of the device, including the tapering sections, of 800 nm. The coupling region of the device is therefore of ultra-compact subwavelength size, since the free-space wavelength λ_0 at the operation frequency of 770 THz is 390 nm.

5.2.2 Ultra Wide-Band Exposure of Human Tissues

Human tissues illuminated by ultra wide-band electromagnetic fields generally exhibit a Debye-type response of the form given by Eq. (37), where the frequencies ω_n are typically determined by experimental and empirical techniques. Table 1 lists typical 2-pole corresponding parameters for the brain, for muscles and for bones [13]. This broadband model is accurate in the frequency range from 100 MHz up to 6 GHz.

Table 1 Parameters for a 2-pole Debye model for wide-band time-domain simulation for different human tissues

Parameter	Brain	Muscles	Bones
$\epsilon_{r,\infty}$	20.20	21.02	5.48
σ_0 in S/m	0.355	0.699	0.104
$\Delta \epsilon_{r,1}$	30.89	12.08	3.19
$\Delta \epsilon_{r,2}$	24.44	34.20	6.20
f_1 in MHz	164.8	212.8	239.6
f_2 in GHz	9.086	11.737	6.707

Recently, specific emphasis has been put on ultra wide-band (UWB) imaging techniques for medical diagnosis and biological detection, which are based, in contrast to X-ray imaging, on low power and non-ionizing radiation while still providing high resolution. These techniques require not only exact dispersive models but also broadband analysis capability. Moreover, the complexity of the materials to investigate is such that an explicit time domain technique with no matrix inversion is the method of choice. In this context, time-domain methods such as the proposed EC FDTD technique may play a crucial role, since they allow almost unlimited description of the geometrical and dispersive properties of these complex materials.

6 Conclusion

The EC FDTD method has been derived for the simplest case of non-dispersive materials and for the case of dispersive materials, including Drude, Lorentz and Debye dispersions.

Within the framework of the EC FDTD a discretized and circuital version of the Poynting theorem both for the time-continuous and time-discrete cases was

established. In the former case the theorem has exactly the same form as the well-known one in the space-and-time-continuous world. Whereas the time-discrete theorem with its additional power flow terms could be utilized to derive novel stability criteria based on a Lyapunov energy function. With the help of this generalized energy function stability criteria have been derived for the non-dispersive, the ε -Drude-dispersive and the double-Drude-dispersive case. At least for these two dispersive materials the maximal possible time iteration step was decreased compared to a non-dispersive media due to additional reactances in the 3D equivalent circuit.

Finally several application examples of metamaterial and natural material dispersive structures and devices have been presented. The diversity of the examples indicates the versatility of the proposed EC FDTD method.

By incorporating the divergence information of the electrodynamic field, or in the circuitual world of the EC FDTD the conservation of charges, an even improved stability criterion is expected. This will be investigated in the near future.

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